

Resolving Inconsistencies in Disjunctive Temporal Constraints

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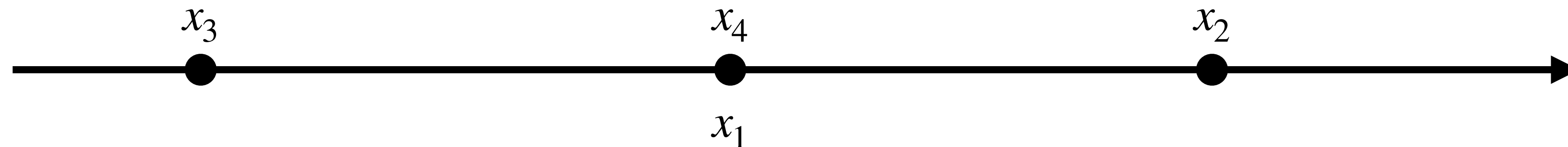
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KR 2026

Temporal Reasoning Formalisms

Objects: points on the timeline $x_1, x_2, \dots, x_n$



Point Algebra: $x_3 < x_1$ $x_1 = x_4$ $x_3 \neq x_2$ $x_3 \leq x_4$

STP: $0 \leq x_1 - x_3 \leq 3$ $0 \leq x_1 - x_4 \leq 0$ $-\infty \leq x_3 - x_2 \leq 10$

2-DTP: $0 \leq x_1 - x_3 \leq 1 \vee 3 \leq x_1 - x_3 \leq 4$

Refs: Vilain, Kautz [AAAI'86], Dechter, Meiri, Pearl [KR'89]

Temporal Reasoning Formalisms

More Formally

Point Algebra (PA): contains binary relations $<, >, \leq, \geq, \neq, =$ over $\mathbb{Q}$.

STP: contains all binary relations $x - y \in [a, b]$ where $a, b \in \mathbb{Q} \cup \{-\infty, +\infty\}$.

2-DTP: contains all binary relations $x - y \in [a_1, b_1] \cup \dots \cup [a_\ell, b_\ell]$.

Checking Consistency

CSP(Γ)

// **c**onstraint **s**atisfaction **p**roblem

In: variables $X = (x_1, \dots, x_n)$,
constraints of the form $x_i R x_j$ for some $x_i, x_j \in X$ and $R \in \Gamma$

Out: is there an assignment $\alpha : X \rightarrow \mathbb{Q}$ that satisfies all constraints?

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Example 1: CSP(PA) instance $x_3 < x_1$, $x_1 = x_4$, $x_2 \neq x_3$, $x_3 \leq x_4$

Assignment $\alpha(x_1) = 2$, $\alpha(x_2) = 3$, $\alpha(x_3) = 1$, $\alpha(x_4) = 3$ satisfies all constraints.

Checking Consistency

CSP(Γ)

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In: variables $X = (x_1, \dots, x_n)$,
constraints of the form $x_i R x_j$ for some $x_i, x_j \in X$ and $R \in \Gamma$

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Example 2: CSP(STP) instance $1 \leq x_1 - x_2 \leq 2, 2 \leq x_2 - x_3 \leq 3, -\infty \leq x_1 - x_3 \leq 2$

Unsatisfiable

Fixing a Few Errors

MinCSP(Γ)

In: instance (X, C) of CSP(Γ) and budget $k \in \mathbb{N}$.

Out: is there $F \subseteq C$, $|F| \leq k$ such that $(X, C \setminus F)$ is satisfiable?

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Example 3: CSP(STP) instance $1 \leq x_1 - x_2 \leq 2$, $2 \leq x_2 - x_3 \leq 3$, $-\infty \leq x_1 - x_3 \leq 2$

No solution for $k = 0$.

Fixing a Few Errors

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Example 3: CSP(STP) instance $1 \leq x_1 - x_2 \leq 2$, $2 \leq x_2 - x_3 \leq 3$, ~~$-\infty \leq x_1 - x_3 \leq 2$~~

No solution for $k = 0$.

Solution for $k = 1$: $F = \{-\infty \leq x_1 - x_3 \leq 2\}$

Computational Complexity

	CSP	MinCSP
Point Algebra	in P	NP-hard
STP	in P	NP-hard
2-DTP	NP-hard	NP-hard

Fine-Grained Analysis

Restricting Expressive Power

$\Gamma \in \{PA, STP, 2-DTP\}$

Question: What is the complexity of $\text{MinCSP}(\Gamma')$ for all $\Gamma' \subseteq \Gamma$?

Fine-Grained Analysis

Restricting Expressive Power

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Question: What is the complexity of $\text{MinCSP}(\Gamma')$ for all $\Gamma' \subseteq \Gamma$?

Answer: $\text{MinCSP}(\Gamma')$ remains NP-hard except for “trivial” cases.



Parameterized Complexity

MinCSP(Γ)

In: instance (X, C) of CSP(Γ) and budget $k \in \mathbb{N}$.

Out: is there $F \subseteq C$, $|F| \leq k$ such that $(X, C \setminus F)$ is satisfiable?

Can we at least solve MinCSP(Γ) when k is small?

Parameterized Complexity

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Can we at least solve MinCSP(Γ) when k is small?

pNP-h: CSP(Γ) is NP-hard (hard even for $k = 0$)

XP: if CSP(Γ) is in P, try all k -subsets F . Requires $|C|^{O(k)}$ time.

FPT: can we solve it in $f(k) \cdot |C|^{O(1)}$ time? // $f(k)$ could be $2^k, 3^k, k!, 2^{2^k}$

W[1]-h: as hard as finding a k -Clique (strong evidence $\notin$ FPT)

Parameterized Complexity Classification

Restricting Expressive Power

$\Gamma \in \{PA, STP, 2-DTP\}$

Question: What is the parameterized complexity of $\text{MinCSP}(\Gamma')$ for all $\Gamma' \subseteq \Gamma$?

- [OPW ESA'23] classified $\text{MinCSP}(\Gamma')$ for all $\Gamma' \subseteq PA$.
- [DJO AAI'22] classified $\text{MinCSP}(\Gamma')$ for all $\Gamma' \subseteq STP$.
- This work: classified $\text{MinCSP}(\Gamma')$ for all $\Gamma' \subseteq 2-DTP$.

Classification for 2-DTP

Recall: *2-DTP* relations $x - y \in [a_1, b_1] \cup \dots \cup [a_\ell, b_\ell]$ where $a_i, b_i \in \mathbb{Q} \cup \{-\infty, +\infty\}$

Question: What is the parameterized complexity of $\text{MinCSP}(\Gamma')$ for all $\Gamma' \subseteq 2\text{-DTP}$?

Classification for 2-DTP

WLOG

Recall: 2-DTP relations $x - y \in [a_1, b_1] \cup \dots \cup [a_\ell, b_\ell]$ where $a_i, b_i \in \mathbb{Z} \cup \{-\infty, +\infty\}$

Question: What is the parameterized complexity of $\text{MinCSP}(\Gamma')$ for all $\Gamma' \subseteq 2\text{-DTP}$?

Classification Theorem

Based on [BMM JACM'18]

2-DTP relations $x - y \in [a_1, b_1] \cup \dots \cup [a_\ell, b_\ell]$ where $a_i, b_i \in \mathbb{Z} \cup \{-\infty, +\infty\}$

For every $\Gamma' \subseteq 2\text{-DTP}$, there exists B such that $\text{CSP}(\Gamma') = \text{CSP}(B)$ and OOTFH:

1. B has a finite domain
2. $B \subseteq PA$
3. $B \subseteq STP$
4. B is a set of periodic 2-DTP relations
5. B is a set of equations and disequations
6. $\text{CSP}(B)$ is NP-hard

Classification Theorem

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Classification Theorem

2-DTP relations $x - y \in [a, b]$

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Finite Domain Case

Example

$$|x_1 - x_2| = 1 \approx x_1 - x_2 \in \{-1, -1\} \cup \{1, 1\}$$

$$|x_2 - x_3| = 1$$

$$|x_3 - x_4| = 1$$

Claim: satisfiable over $\mathbb{Z}$ iff satisfiable over $\{0, 1\}$

$$\alpha(x_1) = 1 \quad \alpha(x_2) = 2 \quad \alpha(x_3) = 3 \quad \alpha(x_4) = 4$$

$$\beta(x_1) = 1 \quad \beta(x_2) = 0 \quad \beta(x_3) = 1 \quad \beta(x_4) = 0$$

Classification Theorem

2-DTP relations $x - y \in [a, b]$

For every $\Gamma' \subseteq 2\text{-DTP}$, there exists a structure $\mathbf{B}$ such that

1. $\mathbf{B}$ has a finite domain

2. $\mathbf{B} \subseteq PA$

3. $\mathbf{B} \subseteq STP$

4. $\mathbf{B}$ is a set of periodic 2-l

5. $\mathbf{B}$ is a set of equations a

6. $\text{CSP}(\mathbf{B})$ is NP-hard

Finite Domain Case Classification?

$\mathbf{B}$ has finite domain.

$\text{CSP}(\mathbf{B})$ dichotomy: Bulatov [FOCS'17], Zhuk [JACM'20]

MinCSP($\mathbf{B}$) dichotomy:

- P vs NP-h: Thapper & Zivny [JACM'16]

- FPT vs W[1]-h: wide open

Classification Theorem

2-DTP relations $x - y \in [a_1, b_1] \cup \dots \cup [a_\ell, b_\ell]$ where $a_i, b_i \in \mathbb{Z} \cup \{-\infty, +\infty\}$

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Classification Theorem

2-DTP relations $x - y \in [a, b]$

For every $\Gamma' \subseteq 2\text{-DTP}$, there exists a constraint language B such that

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Point Algebra Case Example

$$x_1 - x_2 \in [1, \infty]$$

$$x_1 > x_2$$

$$x_2 - x_3 \in [0, \infty]$$

$$x_2 \geq x_3$$

$$x_3 - x_4 \in [-\infty, -1] \cup [1, \infty]$$

$$x_3 \neq x_4$$

Classification Theorem

2-DTP relations $x - y \in [a, b]$

For every $\Gamma' \subseteq 2\text{-DTP}$, there exists

1. B has a finite domain

2. $B \subseteq PA$

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5. B is a set of equations and inequalities

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Point Algebra Case Classification

Let **$B \subseteq PA$** .

$CSP(B)$ is in P for all **B** .

$MinCSP(B)$ is either trivial or NP-hard.

$MinCSP(B)$ is FPT unless it contains both $\leq$ and $\neq$,
otherwise W[1]-hard

[OPW ESA'23]

Classification Theorem

2-DTP relations $x - y \in [a_1, b_1] \cup \dots \cup [a_\ell, b_\ell]$ where $a_i, b_i \in \mathbb{Z} \cup \{-\infty, +\infty\}$

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STP Case

STP is 2-DTP without disjunctions

$\text{CSP}(B)$ is in P.

$\text{MinCSP}(B)$ is either trivial or NP-hard.

$\text{MinCSP}(B)$ is mostly W[1]-hard except for two cases:

- all relations are of the form $x - y \in [a, a]$
- all relations are of the form $x - y \in [a, \infty]$ for $a \geq 0$

[DJO AAI'22]

Classification Theorem

2-DTP relations $x - y \in [a_1, b_1] \cup \dots \cup [a_\ell, b_\ell]$ where $a_i, b_i \in \mathbb{Z} \cup \{-\infty, +\infty\}$

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Periodic Case Example

$$x_1 - x_2 \in \{0, 2, 4\}$$

$$x_1 - x_3 \in \{6, 8\}$$

Equivalent to

$$y_1 - y_2 \in [0, 2]$$

$$y_1 - y_3 \in [3, 4]$$

$$x_1 - x_2 \in [0, 0] \cup [2, 2] \cup [4, 4]$$

$$x_1 - x_3 \in [6, 6] \cup [8, 8]$$

Classification Theorem

2-DTP relations $x - y \in [a, b]$

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Periodic Case Classification

$$x_1 - x_2 \in \{0, 2, 4\}$$

$$x_1 - x_3 \in \{6, 8\}$$

$$y_1 - y_2 \in [0, 2]$$

$$y_1 - y_3 \in [3, 4]$$

$$x_1 - x_2 \in [0, 0] \cup [2, 2] \cup [4, 4]$$

$$x_1 - x_3 \in [6, 6] \cup [8, 8]$$

Reduces to STP with finite intervals.

Classification Theorem

2-DTP relations $x - y \in [a_1, b_1] \cup \dots \cup [a_\ell, b_\ell]$ where $a_i, b_i \in \mathbb{Z} \cup \{-\infty, +\infty\}$

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Equations and Disequations

Example

$$x_1 - x_2 = 1$$

$$x_1 - x_3 \neq 2$$

$$x_1 - x_2 \in [1, 1]$$

$$x_1 - x_3 \in [-\infty, 1] \cup [3, \infty]$$

$\text{CSP}(\mathbf{B})$ is in P.

$\text{MinCSP}(\mathbf{B})$ is either trivial or NP-hard.

$\text{MinCSP}(\mathbf{B})$ is FPT!

Classification Theorem

2-DTP relations $x - y \in [a_1, b_1] \cup \dots \cup [a_\ell, b_\ell]$ where $a_i, b_i \in \mathbb{Z} \cup \{-\infty, +\infty\}$

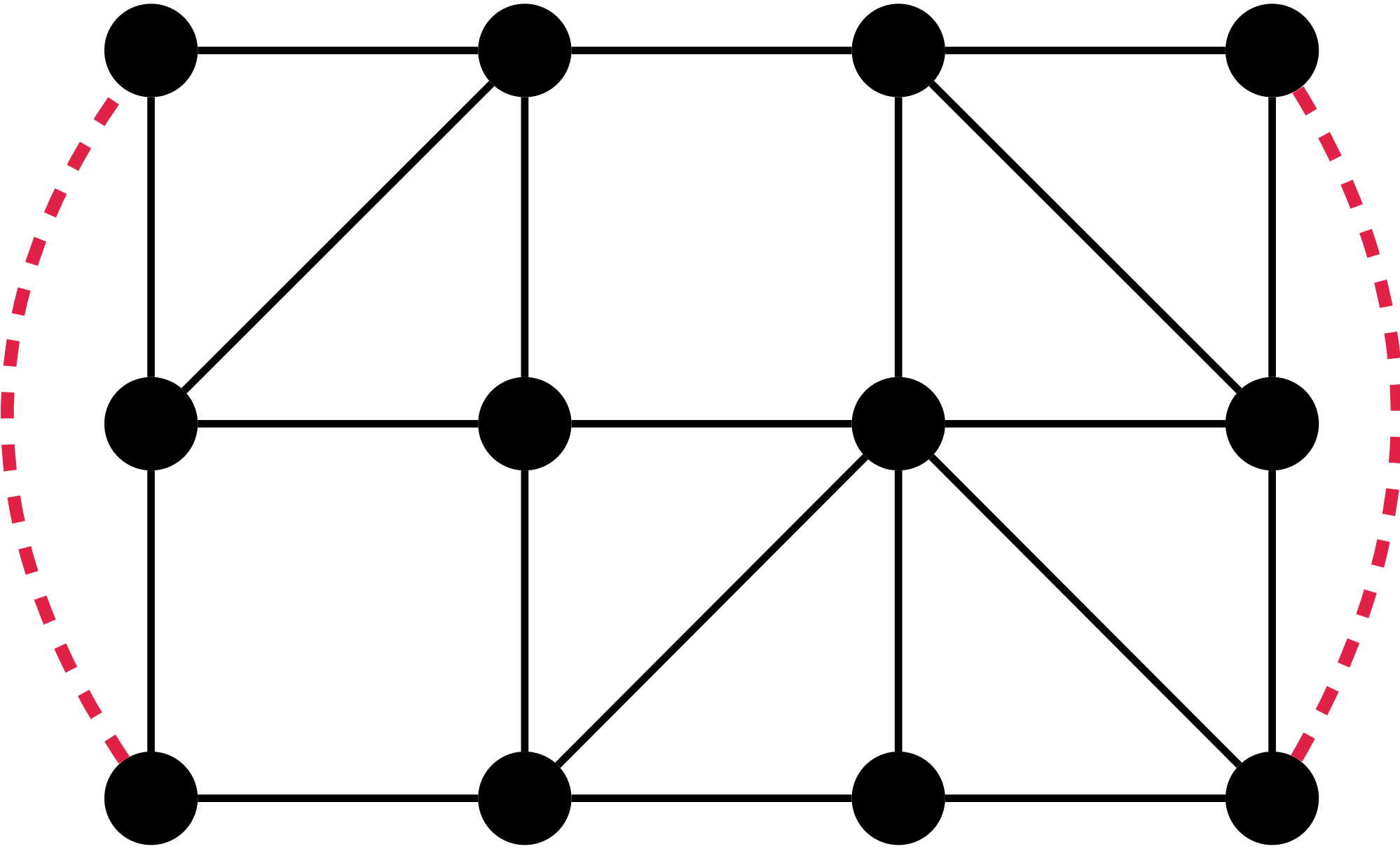
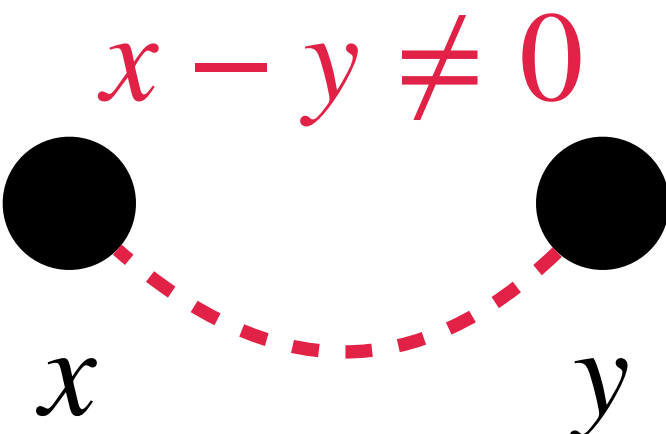
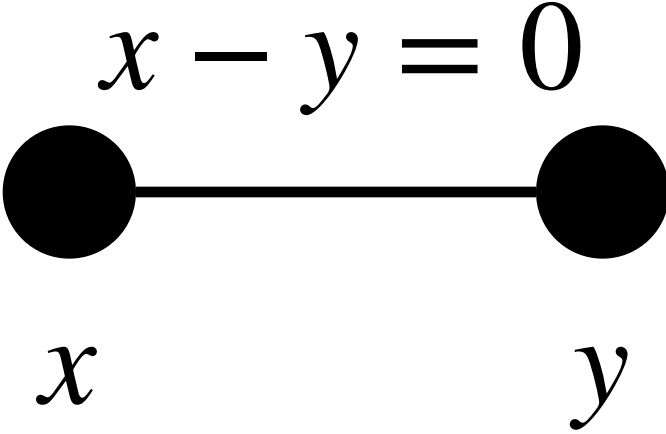
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Equations and Disequations

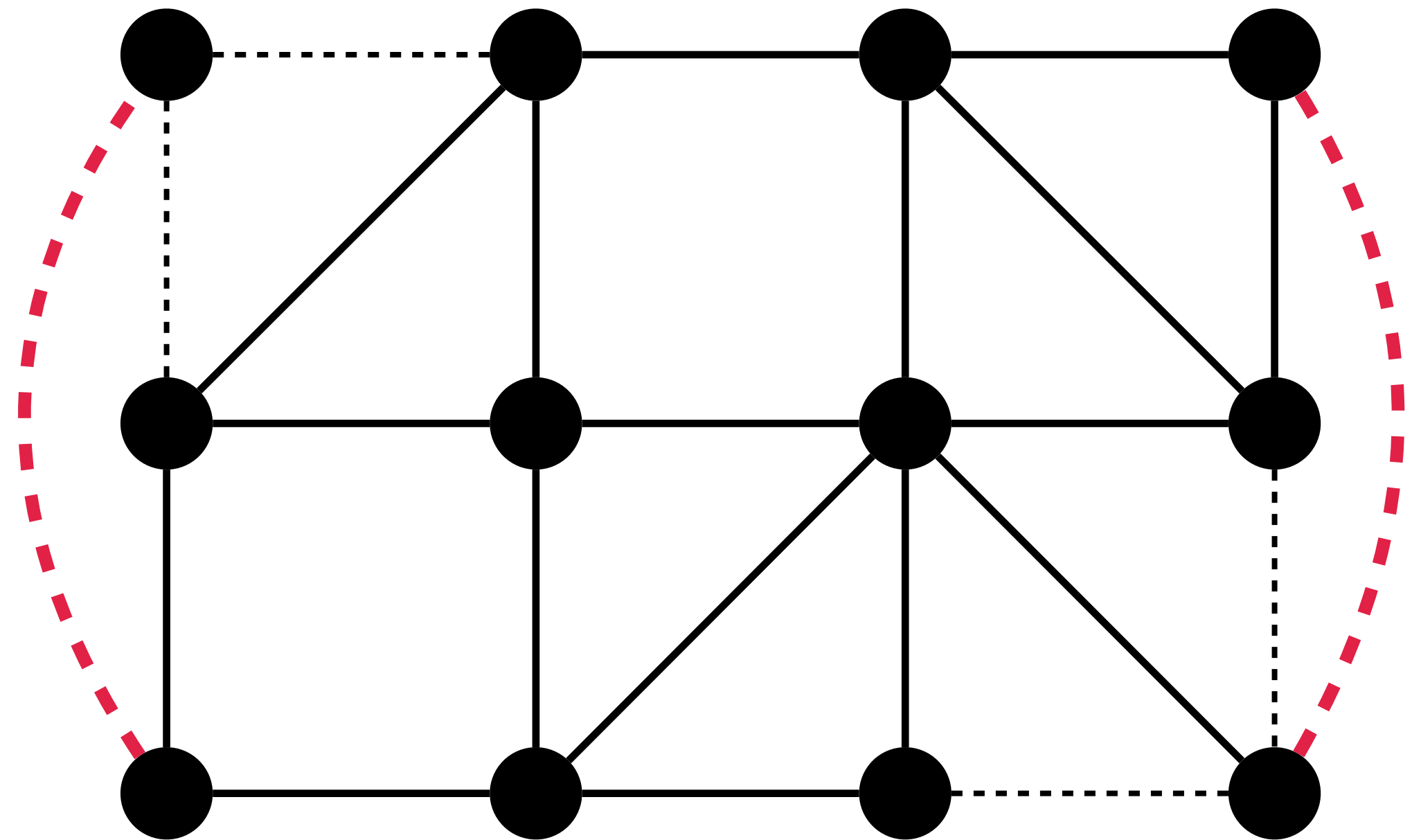
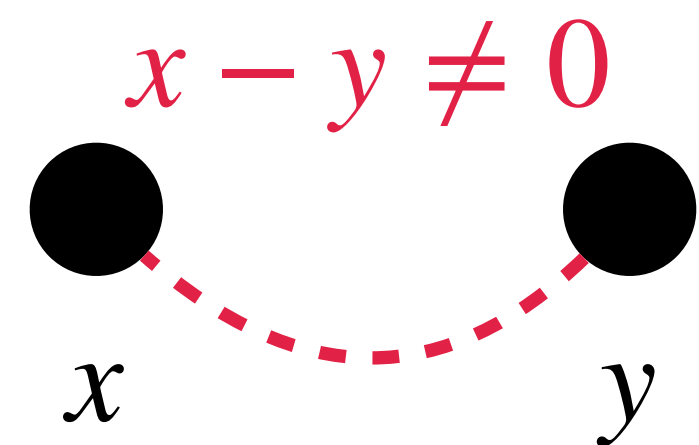
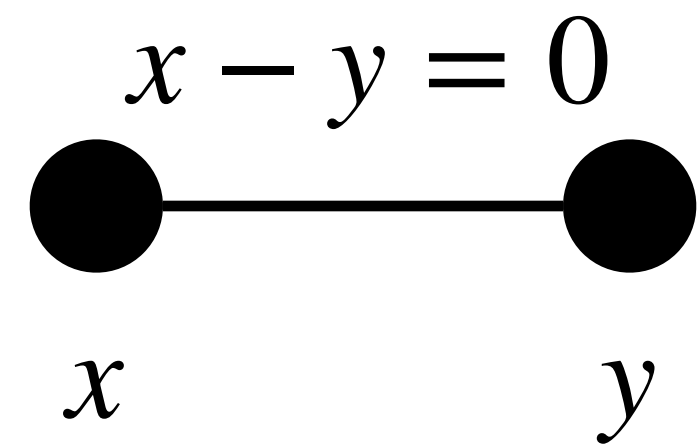
Equations and Disequations

Obstructions of the first kind



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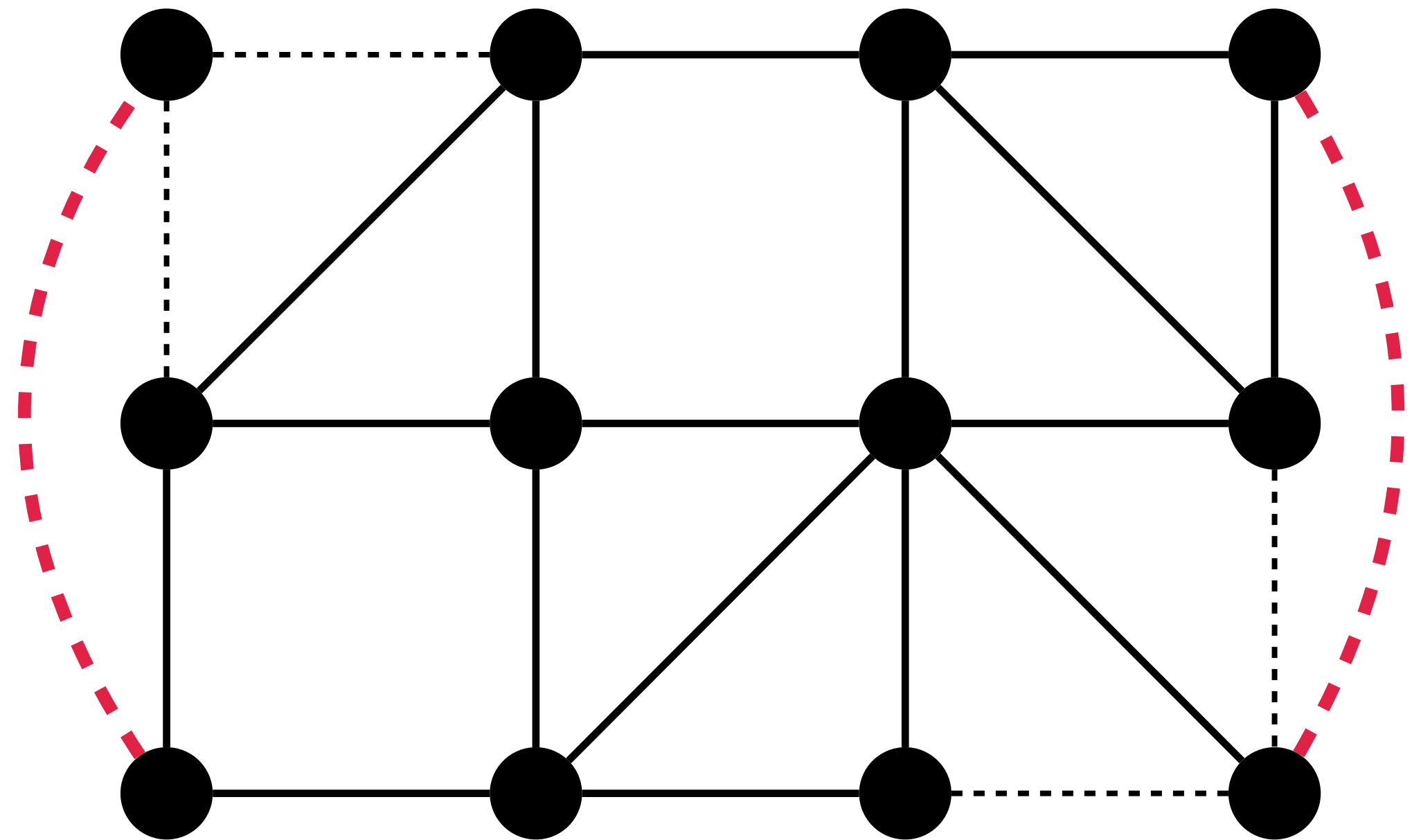
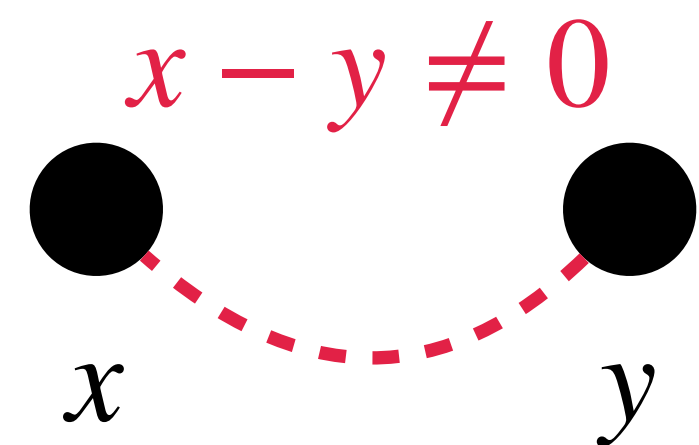
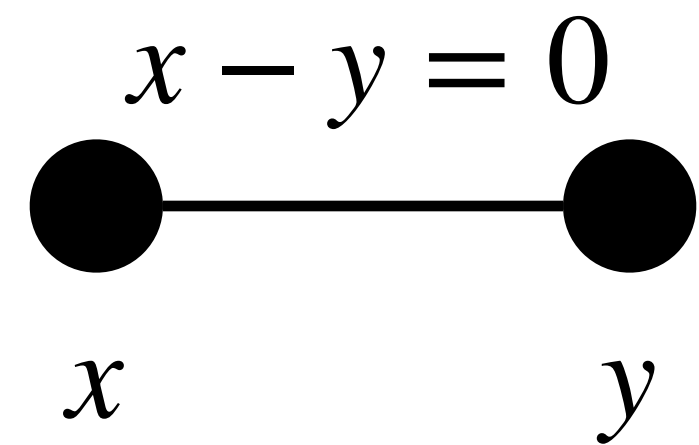
Edge Multicut

In: graph G , cut requests $(s_1, t_1), \dots, (s_m, t_m)$

Out: edge subset X of size $\leq k$ that separates all (s_i, t_i) -pairs

Equations and Disequations

Obstructions of the first kind



Edge Multicut

In: graph G , cut requests $(s_1, t_1), \dots, (s_m, t_m)$

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FPT! [MR SICOMP'14]

Equations and Disequations

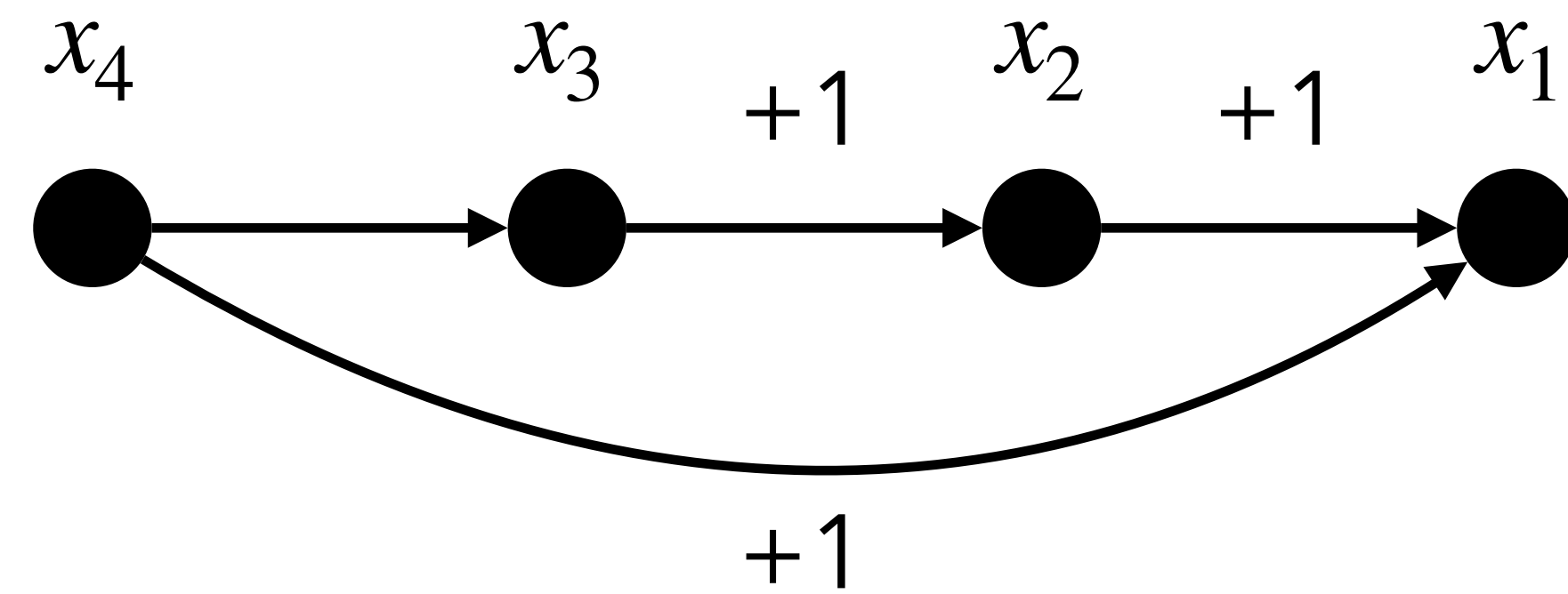
Obstructions of the second kind

$$x_1 - x_2 = 1 \equiv x_2 + 1 = x_1$$

$$x_2 - x_3 = 1 \equiv x_3 + 1 = x_2$$

$$x_3 - x_4 = 1 \equiv x_4 + 1 = x_3$$

$$x_1 - x_4 = 1 \equiv x_4 + 1 = x_1$$



Equations and Disequations

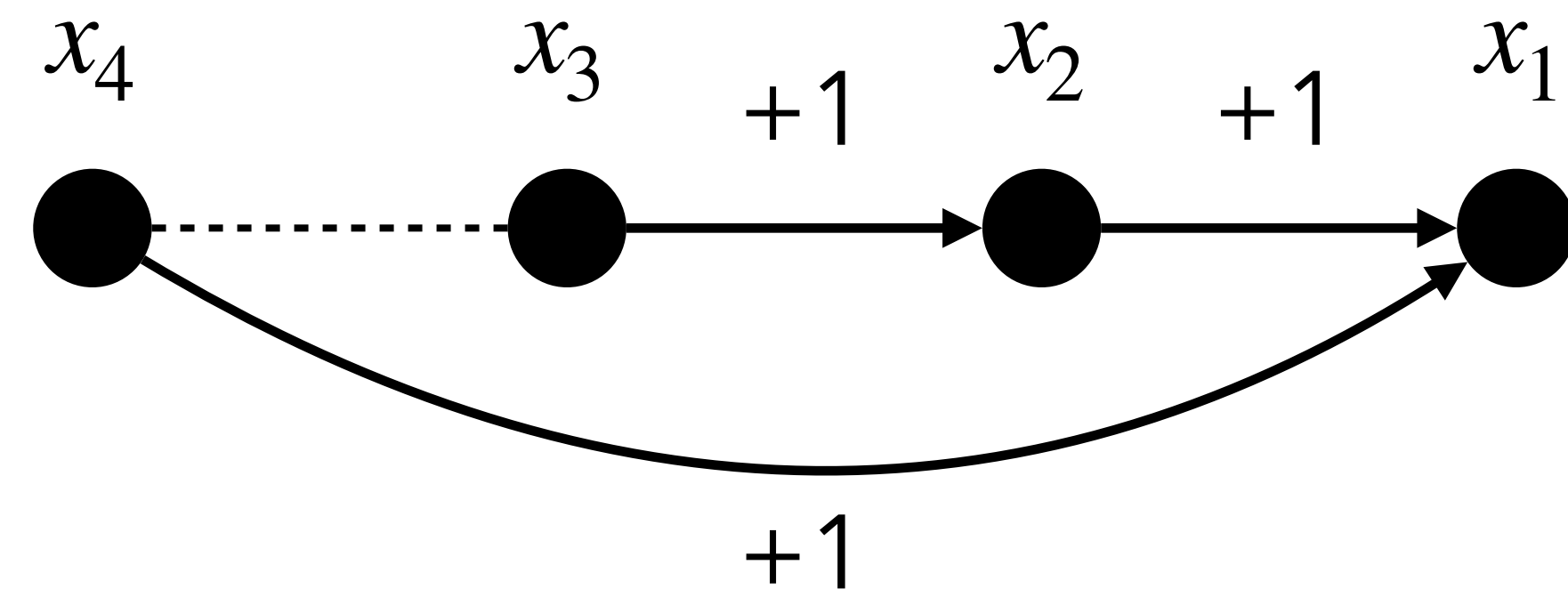
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Group Feedback Set

In: a system of difference equations of the form $x - y = 1$

Out: drop at most k equations to make the system satisfiable

Equations and Disequations

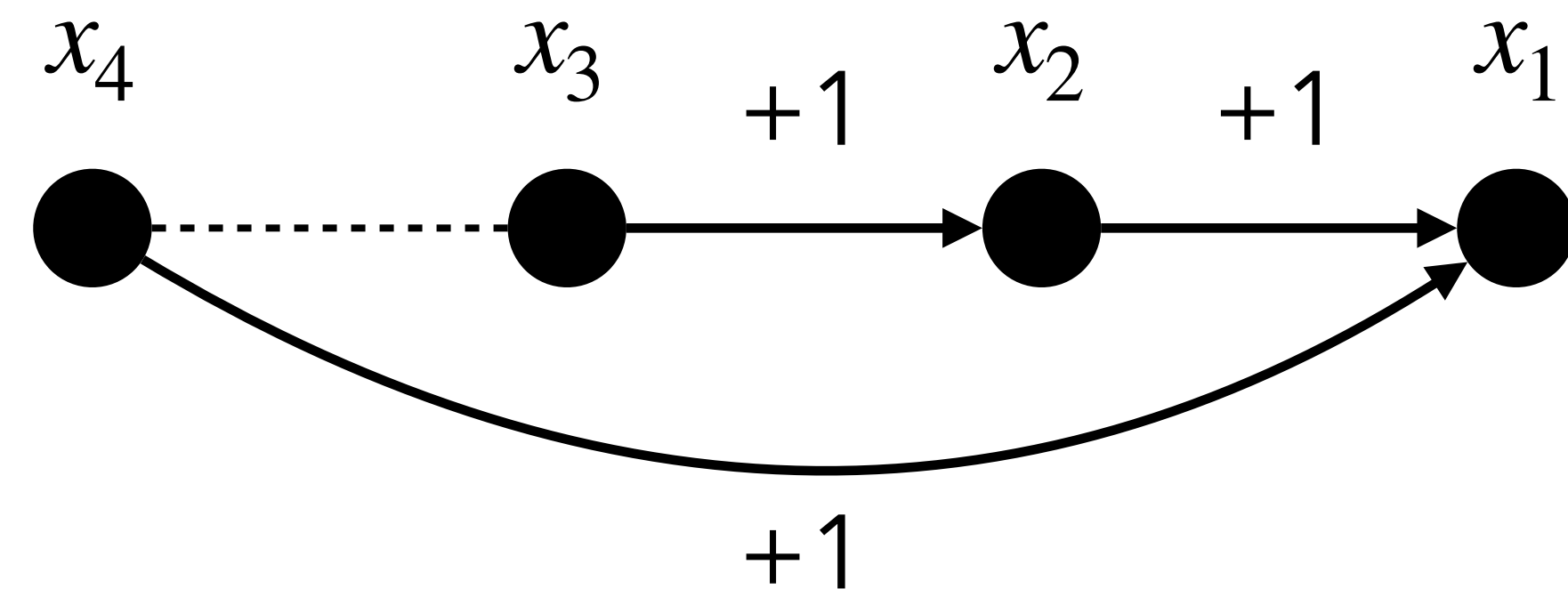
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Group Feedback Set

In: a system of difference equations of the form $x - y = 1$

Out: drop at most k equations to make the system satisfiable

FPT!

[CPP Algorithmic 16]

Equations and Disequations

Obstructions together

A mix of difference equations $x - y = 1$ and disequations $x \neq y$.

One algorithm to solve Edge Multicut and Group Feedback Set.

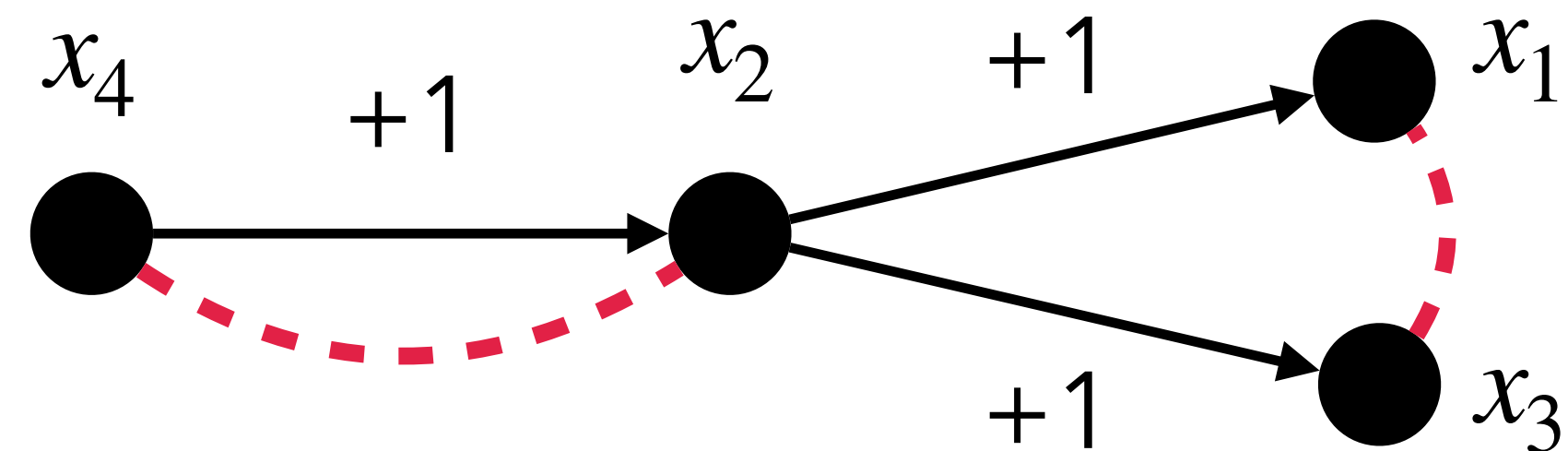
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$$x_3 - x_2 = 1$$

$$x_2 - x_4 = 1$$

$$x_1 \neq x_3$$

$$x_2 \neq x_4$$



Equations and Disequations

Obstructions together

A mix of difference equations $x - y = 1$ and disequations $x \neq y$.

One algorithm to solve Edge Multicut and Group Feedback Set.

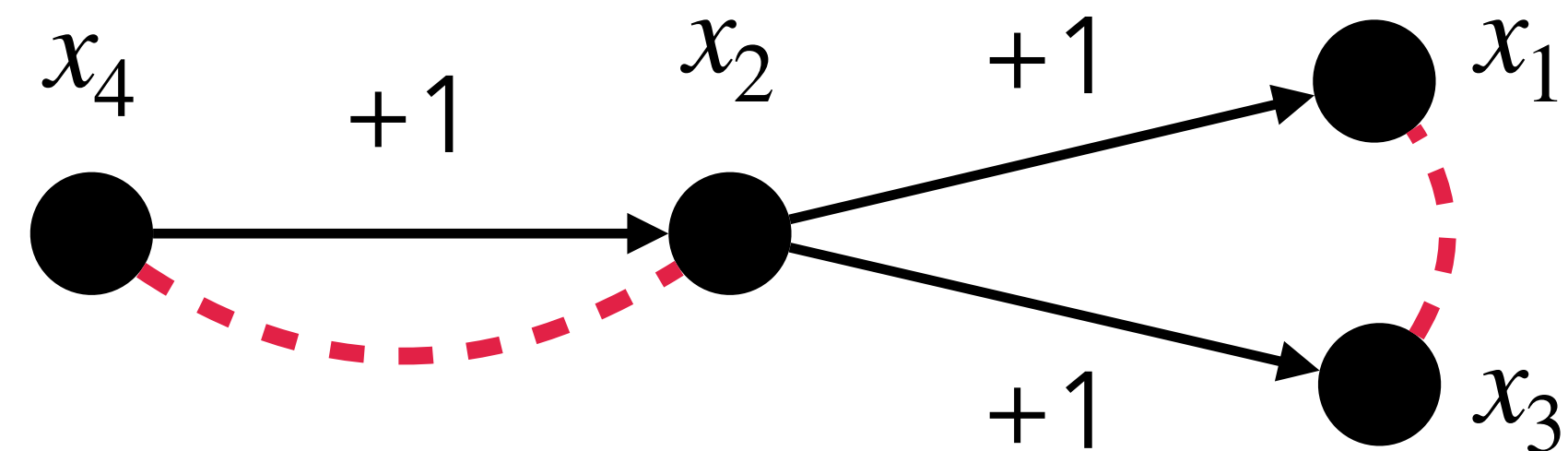
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$$x_1 \neq x_3$$

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Interested in technical details? Come to my talk on Friday (9:30-10:30) at PCCR.

Summary

- **CSP**: consistency-checking problem
- **MinCSP**: resolving *few* inconsistencies
- **Parameterized Complexity**: the right framework for dealing with *few* inconsistencies
- **Fine-Grained Analysis**: study the complexity for all fragments of the formalism

- Point Algebra ✓

- STP ✓

- 2-DTP (this work) ✓

- Future work: temporal CSPs ?

// relations definable using $\exists, \forall, \wedge, \vee, \neg$ over $(\mathbb{Q}; <)$

- Other reasoning formalisms (e.g. spatial) ?